A Brief Intro to Machine Learning

Andres Masegosa

AALBORG UNIVERSITY

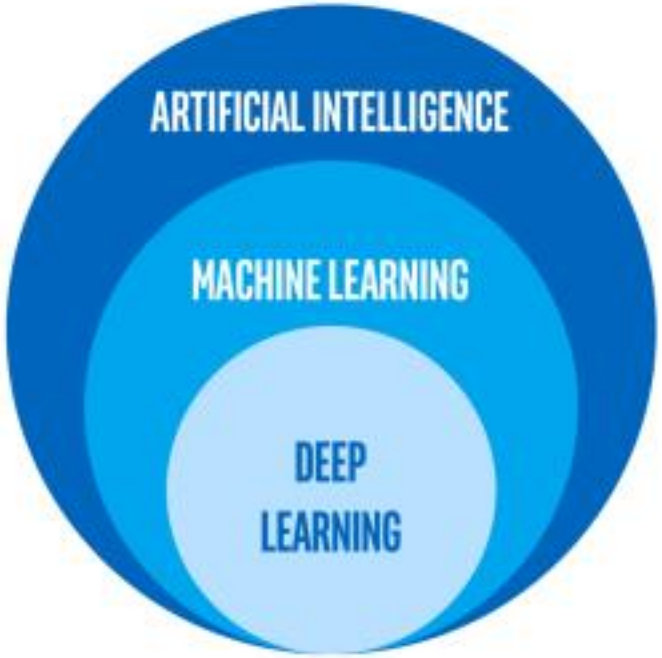
Lecturer - Andrés

- **Research in Machine Learning:**
  - Probabilistic Approach.
  - Robust Machine Learning methods.
  - Probabilistic Programming Languages
  - Theoretical Machine Learning.

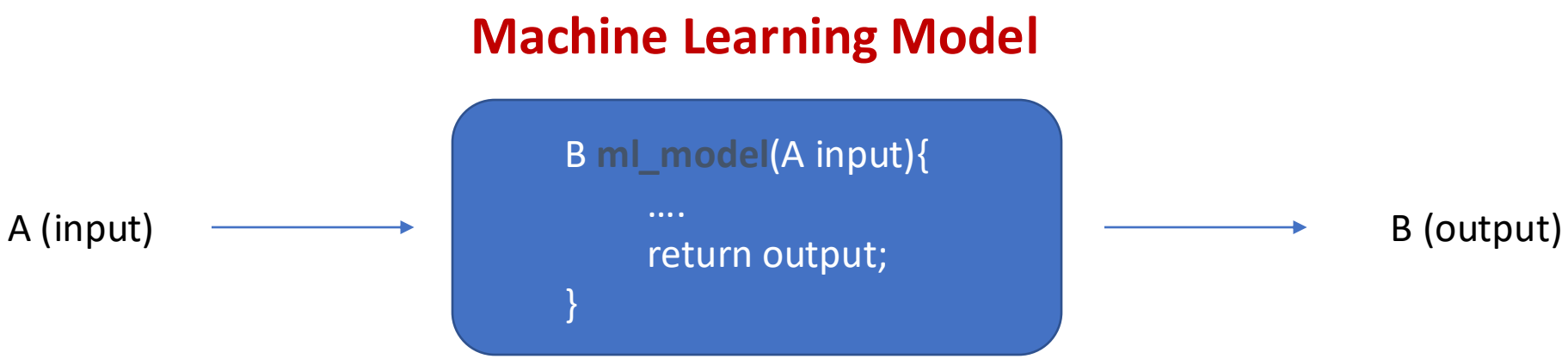
- **Software Development Experience:**
  - Amidst Toolbox (www.amidsttoolbox.com) in Java.
  - InferPy (https://inferpy.readthedocs.io/) in Python.

https://andresmasegosa.github.io/

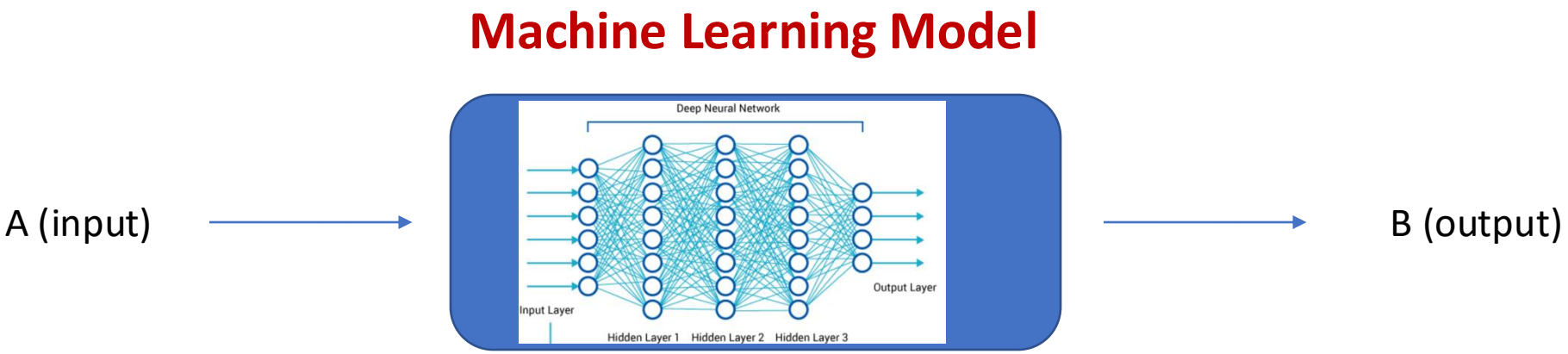
High Level Overview



Machine Learning Model



Machine Learning Model



Machine Learning Model

Spam Filter

Email (String) →

```
B ml_model(A input){
    ....
    return output;
}
```

→ Spam (Boolean)

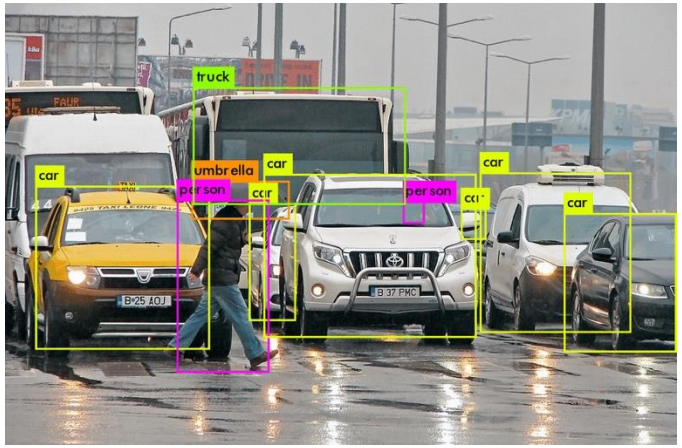
Machine Learning Model

Object Dection

Image (int[][][]) →

```
B ml_model(A input){
    ....
    return output;
}
```

→ Objects (List<String>)



Machine Learning Model

Speech Recognition

Audio (doube[]) →

```
B ml_model(A input){
    ....
    return output;
}
```

→ Transcription (String)

Machine Learning Model

Automatic Translation

Chinese (String) →

```
B ml_model(A input){
    ....
    return output;
}
```

→ English (String)

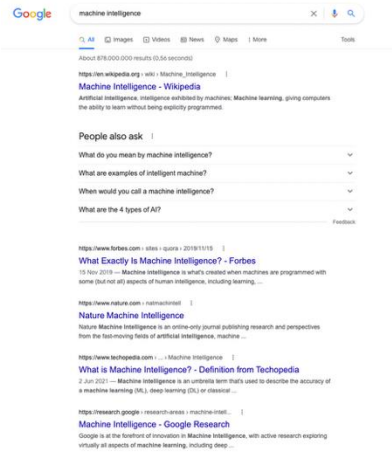
Machine Learning Model

Web Search

Query (String) →

```
B ml_model(A input){
    ....
    return output;
}
```

→ Web Links (List<String>)



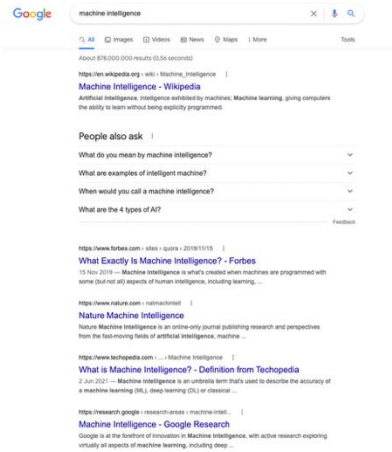
Machine Learning Model

Ad Placement

(Ad, Query, User) →

```
B ml_model(A input){
    ....
    return output;
}
```

→ Click? (boolean)



Machine Learning Model

Self-Driving Cars

(Cameras, LiDar, Sensors, Speed, Brakes, Wheel, etc) →

```
B ml_model(A input){
    ....
    return output;
}
```

→ Car Action (Wheel + Pedals)

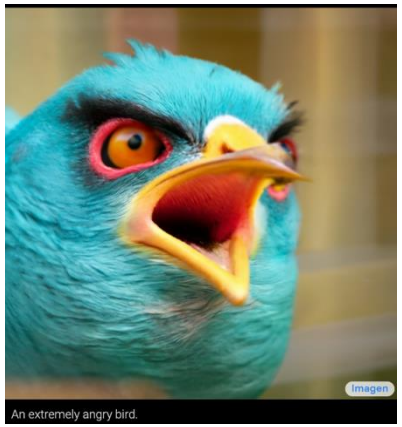
Machine Learning Model

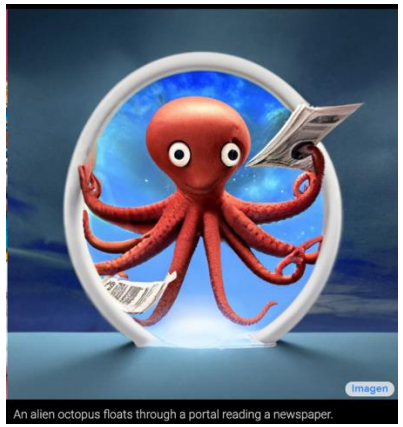
Image Generation

Prompt (Text) →

```
B ml_model(A input){
    ....
    return output;
}
```

→ Image (int[][][])





https://imagen.research.google/

B. *A portrait of a statue of the Egyptian god Anubis wearing aviator goggles, white t-shirt and leather jacket. The city of Los Angeles is in the background. Hi-res DSLR photograph.*

https://parti.research.google/

Anya taylor - joy in the style of gta cover art, by stephen bliss, trending on artstation, pixiv, 8 k

https://lexica.art/

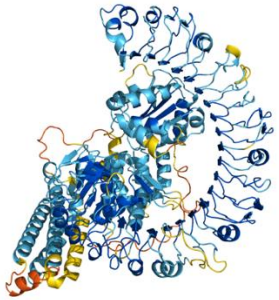
Machine Learning Model

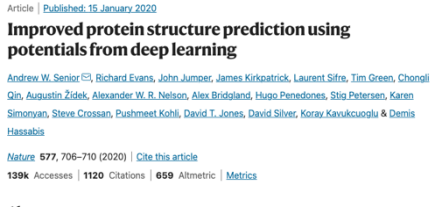
Protein Structure

Gene Sequence (int[])

```
B ml_model(A input){
    ....
    return output;
}
```

Protein Structure





Article | Published: 15 January 2020

Improved protein structure prediction using potentials from deep learning

Andrew W. Senior, Richard Evans, John Jumper, James Kirkpatrick, Laurent Sifre, Tim Green, Chongli Qin, Augustin Žídek, Alexander W. R. Nelson, Alex Bridgland, Hugo Penedones, Stig Petersen, Karen Simonyan, Steve Crossan, Pushmeet Kohli, David T. Jones, David Silver, Koray Kavukcuoglu & Demis Hassabis

Nature **577**, 706–710 (2020) | Cite this article

139k Accesses | **1120** Citations | **659** Altmetric | Metrics

Machine Learning Model

LLMs

Word Sequence (String[])

"There was once upon a"

```
B ml_model(A input){
    ....
    return output;
}
```

Next Word (String)

"time"

Gemini

Exercise (5 min)

ML Application???

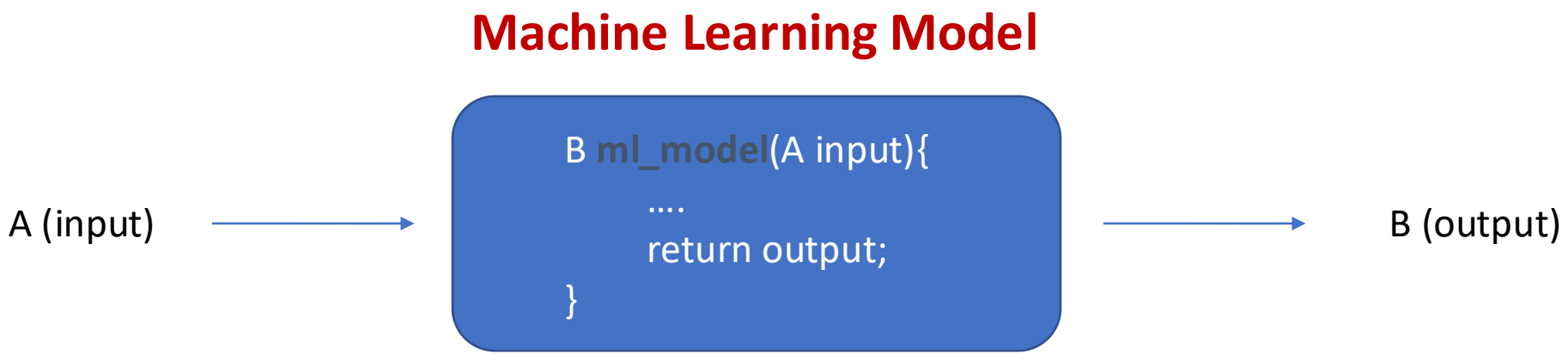
Input??

```
B ml_model(A input){
    ....
    return output;
}
```

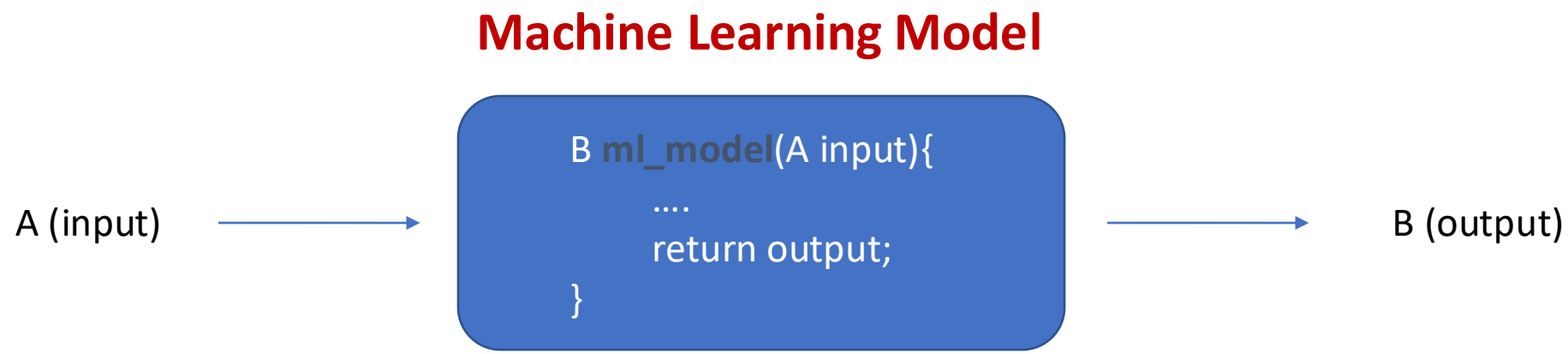
Output??

Think in applications that can be framed in this way

Machine Learning Model



Machine Learning Model



Data is the Fuel of Machine Learning

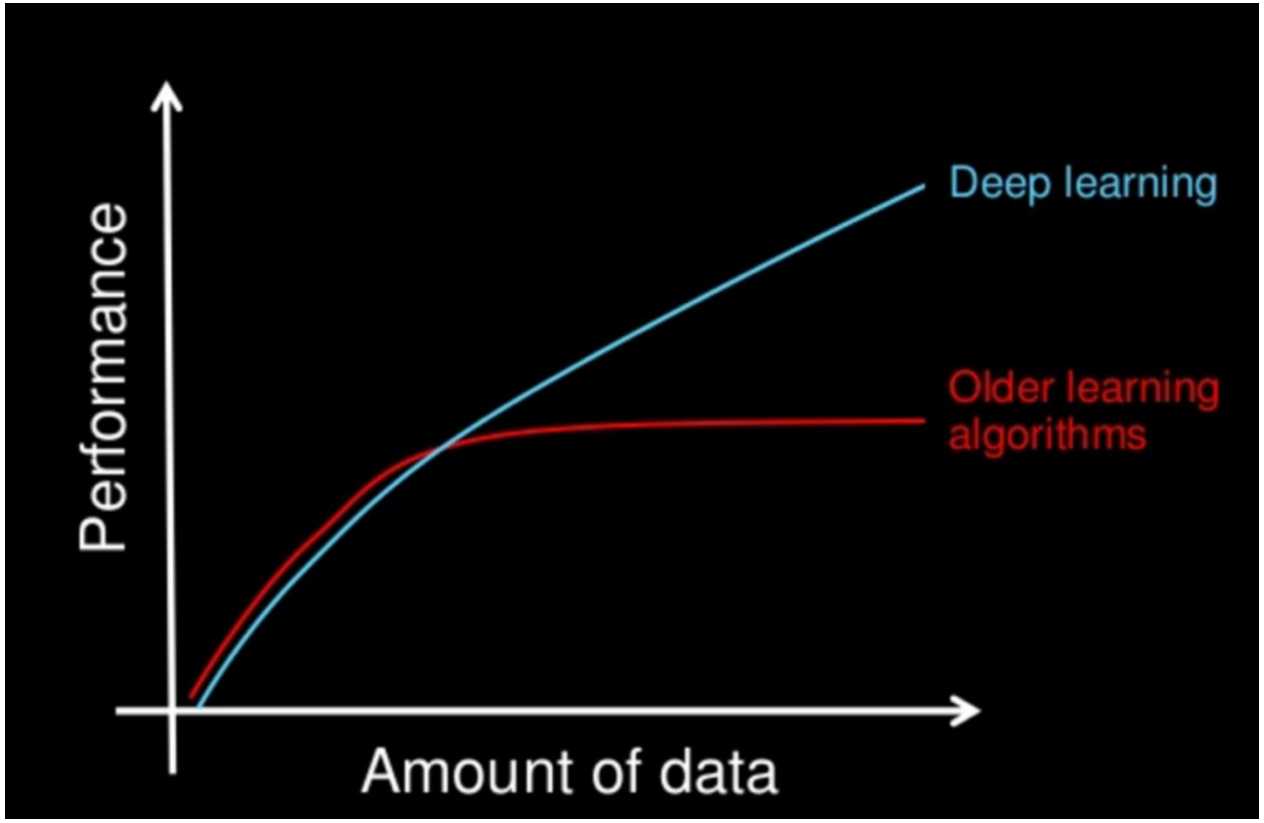
Definition by Tom Mitchell (1998):

Machine Learning is the study of algorithms that

- improve their performance P
- at some task T
- with experience E.

A well-defined learning task is given by <P, T, E>.

Data is the Fuel of Machine Learning



Andrew Ng: Artificial Intelligence is the New Electricity.
https://www.youtube.com/watch?v=21EiKfQYZXc&t=1206s

Data is the Fuel of Machine Learning

Automatic Translation

Data Set

List<(A,B)>

Danish Scandinavian	English
Hus	House
Barn	Child/Baby
Sæbe	Soap
Dag	Day
Gård(by)	Yard(City)
Vi	We
Sin	His/Her
Land(Jord)	Land(Earth)
Bund	Bottom
Modig/ Konge?	Brave/ King?
Knægt/Søn	Boy/ Knight?/Son
Bære	Bear
Bære(Køre)	Carry
Sjæl	Soul
Stellar	Stellar

ML Algorithm

Machine Learning Model

```
B ml_model(A input){
    ....
    return output;
}
```

Data is the Fuel of Machine Learning

Web Search

Data Set

List<(A,B)>

ML Algorithm

Machine Learning Model

```
B ml_model(A input){
    ....
    return output;
}
```

Query	Clicked Web Link
World Cup	https://www.fifa.com/fifaplus/en/tournaments/mens/worldcup/qatar2022
aau software bachelor copenhagen	https://www.aau.dk/uddannelser/bachelor/software
python matrix multiplication	https://numpy.org/doc/stable/reference/generated/numpy.matmul.html
....	....
....	....
....	....
....	....

Data is the Fuel of Machine Learning

Web Search

Data Set

List<(A,B)>

ML Algorithm

Machine Learning Model

```
B ml_model(A input){
    ....
    return output;
}
```

Query	Clicked Web Link
World Cup	https://www.fifa.com/fifaplus/en/tournaments/mens/worldcup/qatar2022
aau software bachelor copenhagen	https://www.aau.dk/uddannelser/bachelor/software
python matrix multiplication	https://numpy.org/doc/stable/reference/generated/numpy.matmul.html
....	....
....	....
....	....
....	....

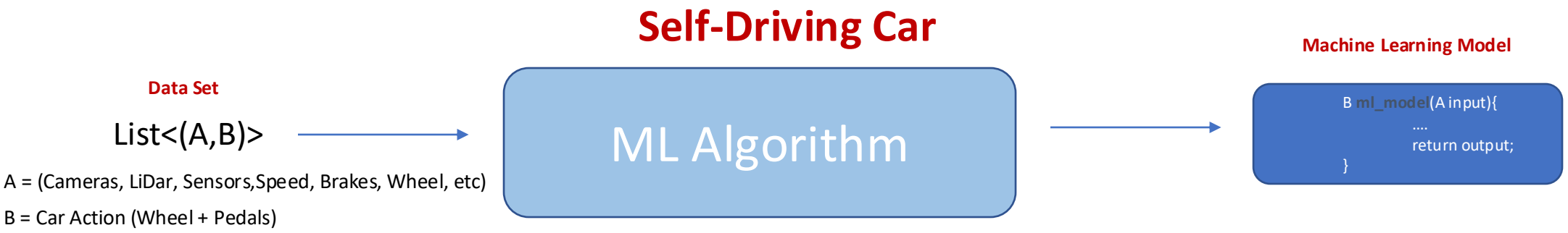
How is Google collecting this data set?

Data is the Fuel of Machine Learning

A = (Cameras, LiDar, Sensors,Speed, Brakes, Wheel, etc)

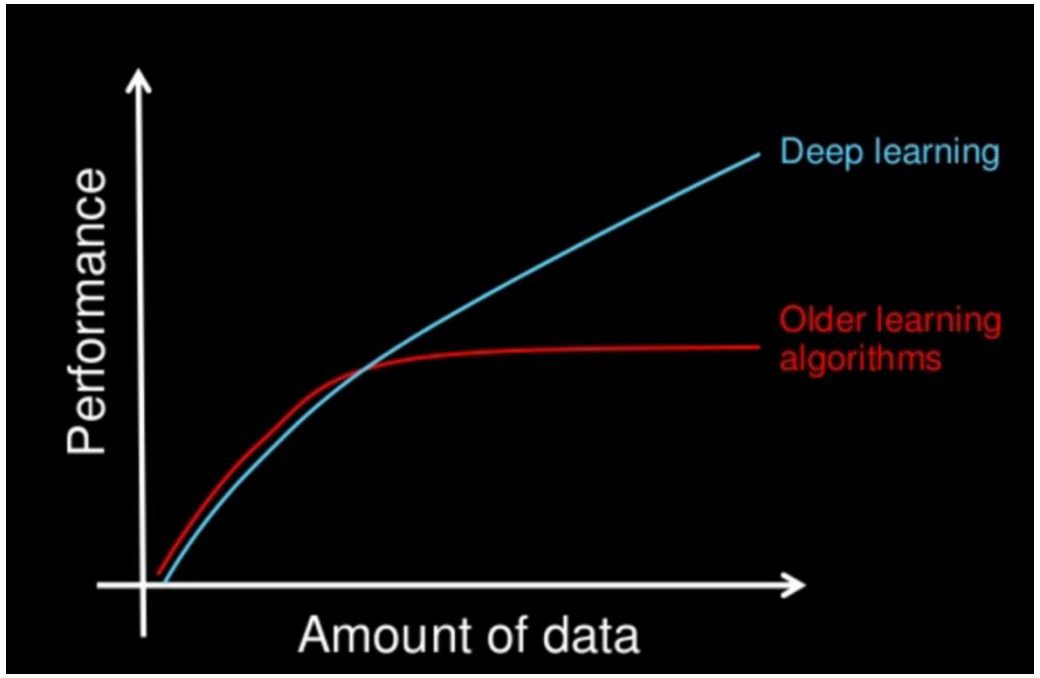
B = Car Action (Wheel + Pedals)

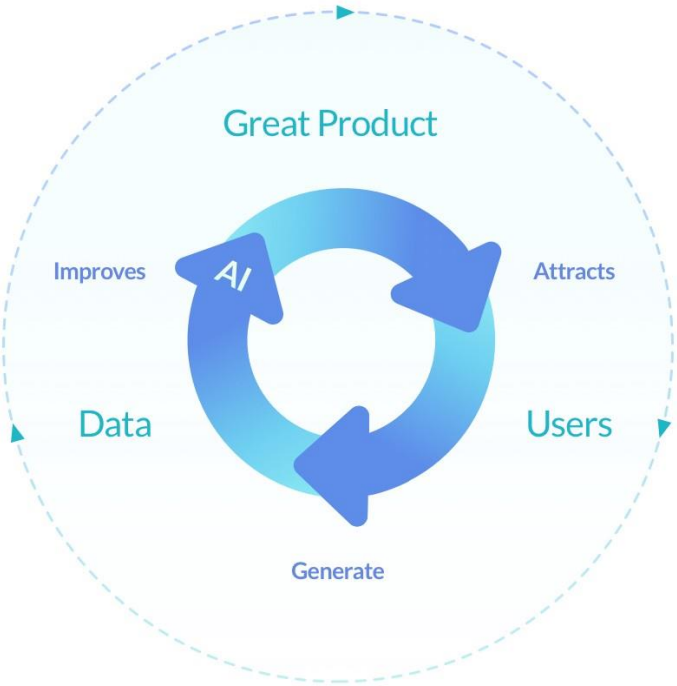
Data is the Fuel of Machine Learning



How is Tesla collecting this data set?

These two figures explain a lot…





Andrew Ng: Artificial Intelligence is the New Electricity.
https://www.youtube.com/watch?v=21EiKfQYZXc&t=1206s

Definition by Tom Mitchell (1998):

Machine Learning is the study of algorithms that

- improve their performance P
- at some task T
- with experience E.

A well-defined learning task is given by <P, T, E>.

Exercise (5 min)

Application???

A (input) →

```
B ml_model(A input){
    ....
    return output;
}
```

→ B (output)

Data Set

List<(A,B)> → ML Algorithm →

Machine Learning Model

```
B ml_model(A input){
    ....
    return output;
}
```

How could you collect data for your application?

BREAK